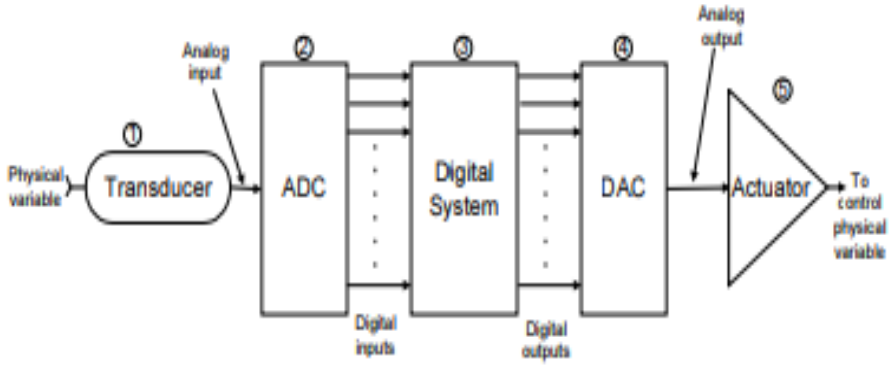
Digital to Analog(DAC) and Analog to Digital (ADC)

Interfacing with analog world

Most physical variables are analog in nature and can take on any value within a continuous range of values. Examples include temperature, pressure, light intensity, audio signals, position, rotational speed, and flow rate. Digital systems perform all of their internal operations using digital circuitry and digital operations. Any information that has to be inputted to a digital system must first be put into digital form. Similarly, the outputs from a digital system are always in digital form.

The physical variable is normally a nonelectrical quantity. A transducer is a device that converts the physical variable to an electrical variable. Some common transducers include thermistors, photocells, photodiodes, flow meters, pressure transducers, and tachometers. The electrical output of the transducer is an analog current or voltage that is proportional to the physical variable it is monitoring. Analog-to digital converter (ADC) and digital-to-converter (DAC) are used to interface a computer to the analog world so that the computer can monitor and control a physical variable .



Digital to Analog Converter (DAC)

Digital output from the computer is connected to a DAC, which converts it to a proportional analog voltage or current. For example, the computer might produce a digital output ranging from 0000000 to 11111111, which the DAC converts to a voltage ranging from 0 to 10V.

Basically, D/A conversion is the process of taking a value represented in digital code (such as straight binary or BCD) and converting it to a voltage or current which is proportional to the digital value. Fig. shows the symbol for a typical 4-bit D/A converter. Now, we will examine the various input/output relationships.

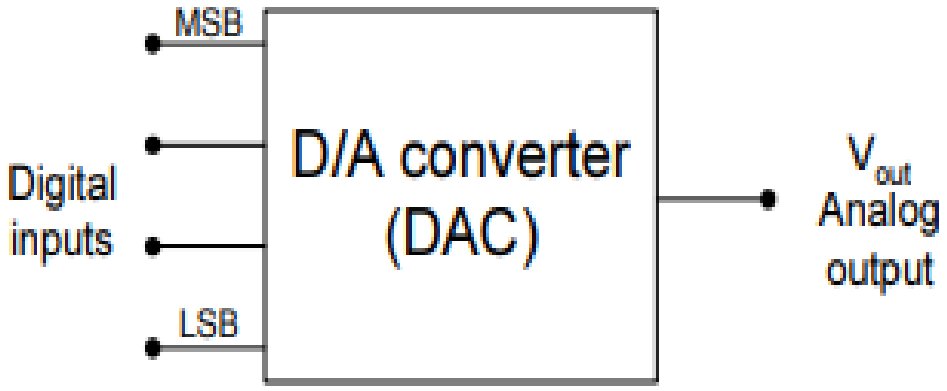


Fig. 7.2 : Four bit DAC with voltage output.

The digital inputs D,C,B, and A are usually derived from the output register of a digital system. The $2^4 = 16$ different binary numbers represented by these 4 bits for each input number, the D/A converter output voltage is a unique value. In fact, for this case, the analog output voltage Vout is equal in volts to the binary number.

In general,

Analog output = K × digital input

where K is the proportionality factor and it is constant value for a given DAC. The analog output can of course be a voltage or current. When it is a voltage, K will be in voltage units, and when the output is current, K will be in current units.

TYPES

The binary weighted resistors produce binary-weighted current which are summed up by the op-amp to produce proportional output voltage. The binary word applied to the switches produces a proportional output voltage. Several different binary codes such as straight binary, BCD and offset binary are commonly used as inputs to D/A converters.

There are several methods and circuits for producing the D/A operation. We shall examine several of the basic schemes, to gain an insight into the ideas used.

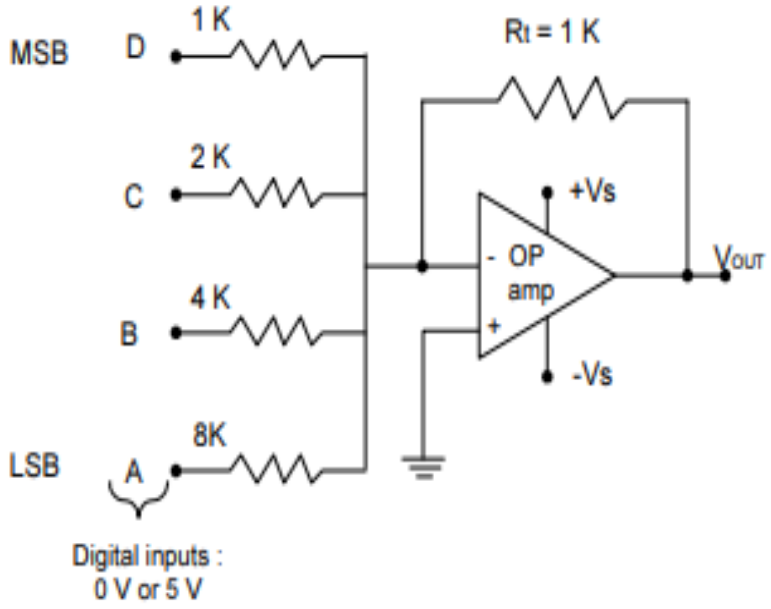


Fig. 7.4 : DAC circuitry using op-amp with binary weighted resistors.

Fig. shows the basic circuit of 4-bit DAC. The inputs A,B,C, and D are binary inputs which are assumed to have values of either 0 V or 5 V. The operational amplifier is employed as a summing amplifier, which produces the weighted sum of these input voltages. the summing amplifier multiplies each input voltage by the ratio of the feedback resistor RF to the corresponding input resistor RIN. In this circuit RF RIN 1kΩ and the input resistors range from 1 to 8 kΩ . The D input has RIN = 1KΩ, so the summing amplifier passes the voltage at D with no attenuation. The C input has RIN = 2 kΩ, so that it will be attenuated by. Similarly, the B input will be attenuated by ¼ and the A input by 1 /8.
The amplifier output can thus be expressed as

VOUT = - (VD +1 /2 VC + ¼ VB + 1 /8 VA)

The negative sign is present because the summing amplifier is a polarity inverting amplifier.

R-2R ladder Network

The DAC circuits we have looked at, has some practical limitations. The biggest problem is the large difference in resistor values between the LSB and MSB, especially in high-resolution DACs. One of the most widely used DAC circuits that uses resistance's fairly close in value is the R/2R ladder network. Here the resistance values span a range of only 2 to 1. Note, how the resistors are arranged, and only two different values are used, R and 2R. The current IOUT depends on the positions of the four switches, and the binary inputs B3B2B1B0 control the states of the switches. This current is allowed to flow through an op-amp current-to-voltage converter to develop VOUT. It can be shown that the value of VOUT is given by the expression.

$$V_{OUT} = \frac{-V_{REF}}{8} \times B .$$

where B is the value of the binary input, which can range from 0000 to 1111

DAC Specifications

- Resolution: Smallest change that can occur in analog output as result of change in digital input
- Accuracy:

Full scale error : maximum deviation of DACs output from ideal value

linearity error: maximum deviation in step size from ideal step size

- Settling time: Time required for DAC output to go from 0 to F. S. as binary input change from all 0's to all 1's
- Offset error

DAC applications

- Control
- Automatic Testing
- Signal reconstruction (DSO, Digital audio and video recording)

Analog to Digital Converter(ADC)

An analog-to-digital converter takes an analog input voltage and after a certain amount of time produces a digital output code which represents the analog input. The A/D conversion process is generally more complex and time-consuming than the D/A process. The techniques that are used provide and insight into what factors determine an ADCs performance.

ADCs allow micro-processor controlled circuits, Arduinos, Raspberry Pi, and other such digital logic circuits to communicate with the real world. Following figure shows block diagram of 8 bit ADC.

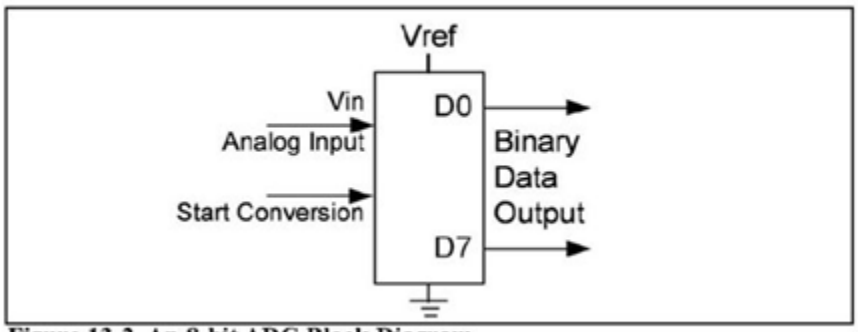


Figure 13-2. An 8-bit ADC Block Diagram

ADC Types:

Type	Max Sample Rate	Resolution (maximum bits)
Flash	10 gigasamples/s	4-12
SAR	10 megasamples/s	8-18
Delta-sigma	1 megasample/s	8-32
Pipelined	1 gigasample/s	8-16

ADC Specifications

1)Conversion Time: It is defined as the total time required for an A/D converter to convert an analog signal to digital output. It depends on the conversion technique and propagation delay of the circuit components.

2)Resolution: The resolution refers to the finest minimum change in the signal which is accepted for conversion, and it is decided with respect to number of bits.

Resolution can also be defined as the ratio of change in the value of input voltage Vi, needed to change the digital output by 1 LSB.

n-bit	Number of steps	Step Size (mV)
8	256	5/256 = 19.53
10	1024	5/1024 = 4.88
12	4096	5/4096 = 1.2
16	65536	5/65536 = 0.076

Notes: $V_{CC} = 5$ V
Step size (resolution) is the smallest change that can be discerned by an ADC.

Table 5.5.1 Resolution Vs Step Size for ADC

3)Analog error: An error occurring due to the variations in DC switching point of the comparator, resistors reference voltage source, ripples and noises introduced by the circuit components is termed as analog error.

4)**Linearity error:** It is defined as the measure of variation in voltage step size. It indicates the difference between the transitions for a minimum step of input voltage change. This is normally specified as fraction of LSB.

Calculation of ADC output

- Digital output= 2^n x Analog input voltage/ref. voltage where n= no. of bits

LM 35 interfacing with ADC 0804

LM35 TEMPERATURE SENSORS: ➢

The LM35 series sensors are precision integrated-circuit temperature sensors whose output voltage is linearly proportional to the celsius (centigrade) temperature.The LM35 requires no external calibration since it is internally calibrated.It outputs 10mV for each degree of centigrade temperature.

Part	Temperature Range	Accuracy	Output Scale
LM35A	–55 C to +150 C	+1.0 C	10 mV/C
LM35	–55 C to +150 C	+1.5 C	10 mV/C
LM35CA	–40 C to +110 C	+1.0 C	10 mV/C
LM35C	–40 C to +110 C	+1.5 C	10 mV/C
LM35D	0 C to +100 C	+2.0 C	10 mV/C

Note: Temperature range is in degrees Celsius.

Table 5.5.3 LM35 Temperature Sensor Series Selection Guide

ADC804 CHIP: ADC804 IC is an 8-bit parallel analog-to-digital converter. It works with +5 volts and has a resolution of 8 bits.

Vref/2: (Pin 9) - It is used for the reference voltage. If this pin is open (not connected), the analog input voltage is in the range of 0 to 5 volts (the same as the Vcc pin). If the analog input range needs to be 0 to 4 volts, Vref/2 is connected to 2 volts.

Table shows the Vin range for various Vref/2 inputs.

$V_{ref}/2$ (V)	V_{in} (V)	Step Size (mV)
not connected*	0 to 5	5/256 = 19.53
2.0	0 to 4	4/255 = 15.62
1.5	0 to 3	3/256 = 11.71
1.28	0 to 2.56	2.56/256 = 10

Notes: V_{CC} = 5 V

*When not connected (open), $V_{ref}/2$ is measured at 2.5 volts for V_{CC} = 5 V.

Step Size (resolution) is the smallest change that can be discerned by an ADC.

Table 5.5.2 Vref/2 Relation to Vin Range (ADC0804)

Since the ADC0804 has 8-bit resolution with a maximum of 256 (2^8) steps and the LM35 (or LM34) produces l0 mV for every degree of temperature change, we can condition Vin of the ADC0804 to produce a Vout, of 2560 mV (2.56 V) for full-scale output.

Therefore, in order to produce the full-scale Vout of 2.56 V for the ADC0804, we need to set Vref/2 = 1.28

To calculate the output voltage, use the following *formula*

Dout =Vin/Step Size

Where Dout = digital data output (in decimal), Vin = analog voltage, and Step size (resolution) is the smallest change, which is (2 * Vref/2)/256 for ADC 0804